

Checkpoint 3: Baseline U-Net Diffusion Model

Objective

Train and evaluate the baseline conditional U-Net denoising backbone.

Key Message

The baseline model gives the reference point for all later comparisons.

Completed Work

- Implemented conditional U-Net with timestep conditioning.
- Completed baseline training and evaluation on the fixed test split.
- Generated metric CSV files and qualitative comparison grids.

Evidence

Metric	Baseline 128/50
MSE	0.039444
PSNR	15.681501
SSIM	0.610758
Delta E	29.645203



Baseline qualitative test examples.

Notes

- This model is the default U-Net diffusion baseline, not the proposed contribution.